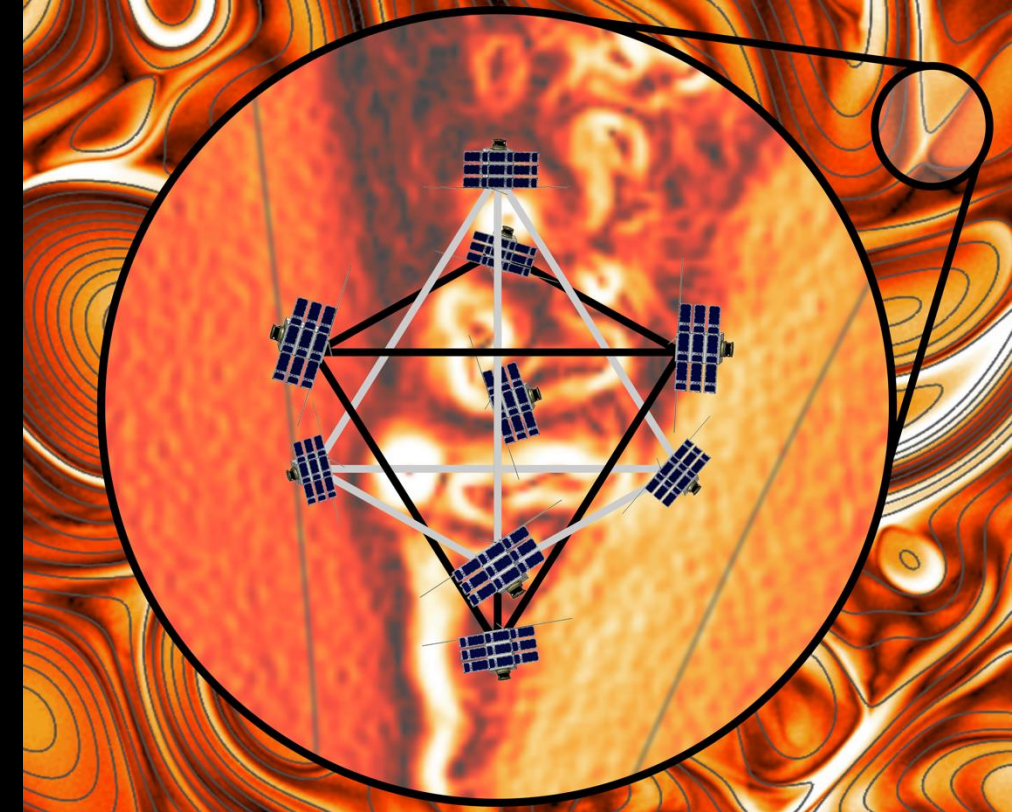


DREAM (Distributed Receivers for Electron Astrophysics Measurements) A Small Satellite Mission Concept for Probing Electron-Scale Energisation Processes

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1 SUMMARY

Understanding **particle energisation** due to plasma dynamics at length scales down to the **smallest electron scales** remains a key open challenge in the field of heliophysics

DREAM is a small satellite swarm concept that would form a distributed electromagnetic field and electron instrument to **quantify electron energisation** and **characterise electron-scale fluctuations** in the solar wind and Earth's magnetosheath

MISSION OVERVIEW

Spacecraft	4 Small Satellites or up to 11 smaller 16U CubeSats 1x: Magnetometer ; 1x: Full-Sky Electron Camera ; up to 8x: Field-Particle Correlation Sensor
Orbit	High-elliptical orbit (apogee > 14 R _E) covering solar wind and Earth's magnetosheath
Formation	Loosely controlled 3D formation with separations from 0.5 – 10km around apogee

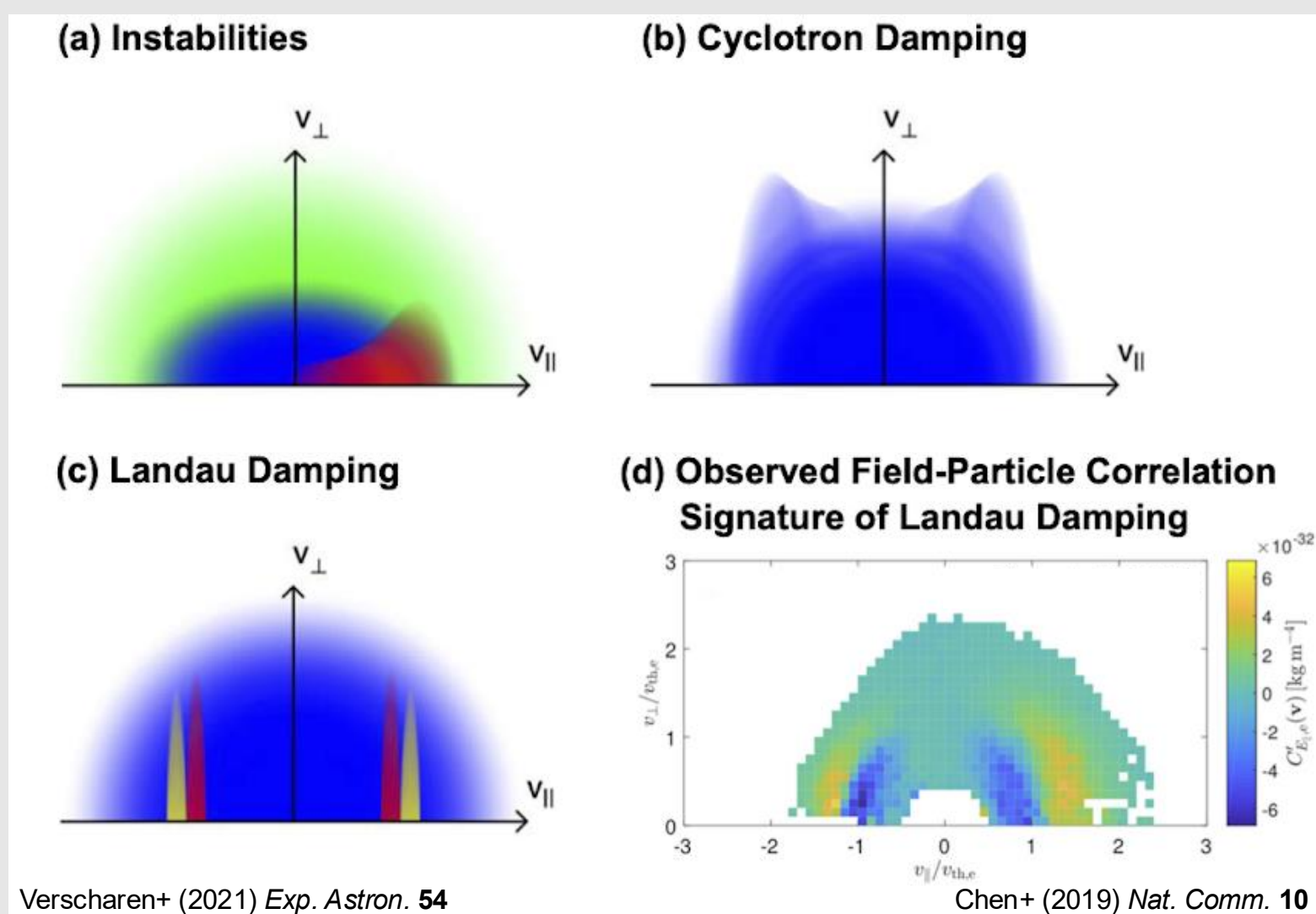
2 ELECTRON ENERGISATION IN COLLISIONLESS PLASMAS

Electron-kinetic processes are the final, crucial step in the dissipation of highly-complex turbulent dynamics prolific in space and astrophysical systems throughout the Universe.

A key question in the study of collisionless plasmas is understanding:

What processes control the energisation and thermodynamics of electrons?

FIGURE 1



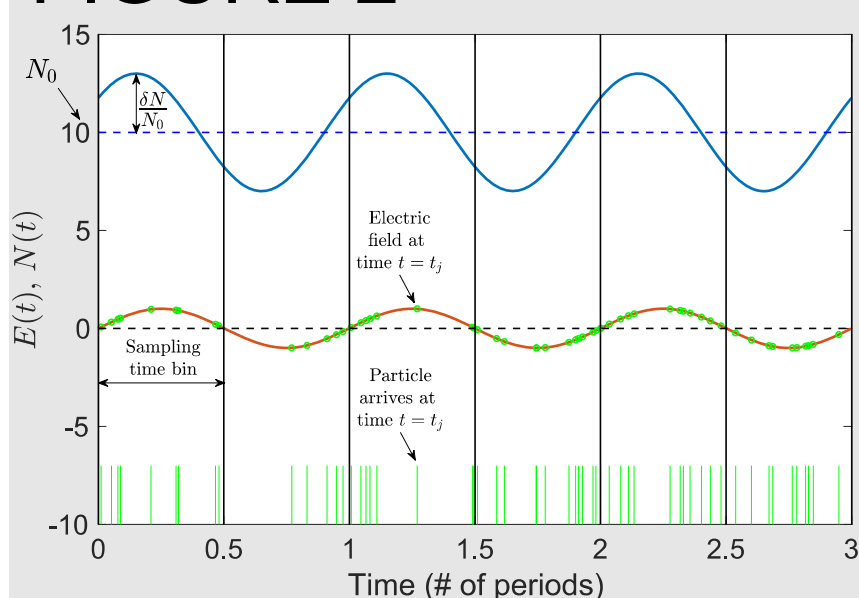
Many processes contribute to regulating electron energisation in collisionless plasmas including instabilities, resonant processes, and nonlinear structures – all of which can be associated with different structure in the velocity distribution function (**Fig. 1a-c**).

Field-Particle Correlations (FPC; Fig. 1d) are a powerful tool for quantifying the amount of electron energisation (integrated signal), determining which particles are energised (velocity space FPC profile), and distinguishing different energisation processes (velocity space FPC profile). [Klein & Howes (2016) *ApJ* **826**; Howes+ (2017) *JPP* **83**; Klein+ (2017) *JPP* **83**]

DREAM will provide a novel dataset of electron field-particle correlations computed on-board the spacecraft

Computing Field-Particle Correlations On-Board a Spacecraft

FIGURE 2



Method 1: Non-uniform time series of electric field at times when electron counts are registered is averaged over time for each velocity-space bin (**Fig. 2**) [Verniero+ (2021) *JGR* **126**].

Method 2: Locking onto phase of intense narrow-band waves with electronics and binning particle counts according wave phase [Kietzing+ (2005) *GRL* **32**].

3 QUANTIFYING ELECTRON ENERGIZATION

Two compact integrated sensor types & a data processing unit form a combined **Field-Particle Correlation Sensor**:

- Magnetic Electron Sensor** – Electron counts and velocities from single look direction
- Electric Field Instrument** – Electric field along the look direction
- Wave-Particle Correlator** – Data processing unit providing interface between sensors

Magnetic Electron Sensor

uses magnetic deflections to separate electrons of different speeds (**Fig. 3**):

- Linear velocity sampling across energy range (*Fine-scale velocity resolution*)
- Simultaneously measure full range of velocities (*Fast correlations with electric field across full energy range*)

****See Poster 16 by Ben Dobbing for more details about magnetic electron sensor****

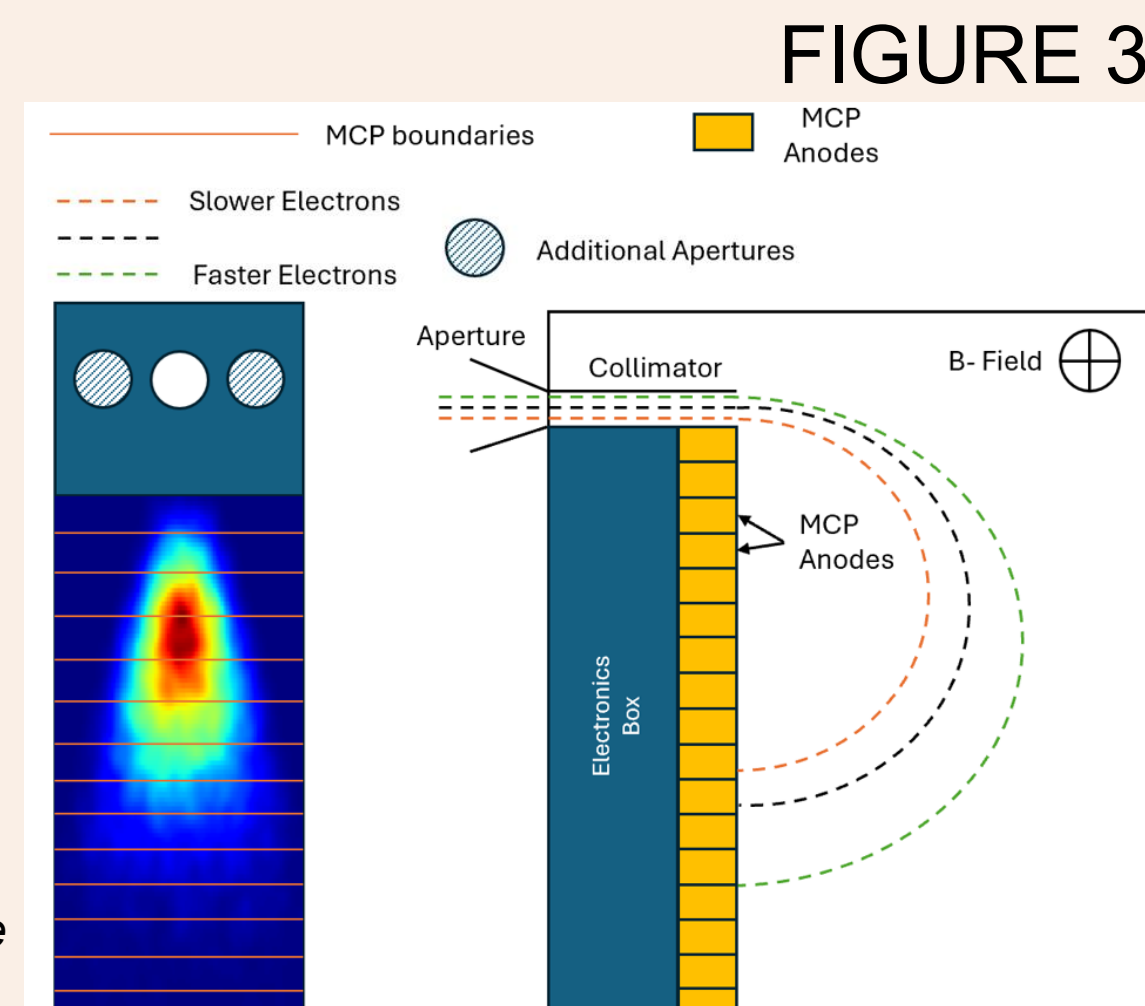


FIGURE 3

Multiple combined **Field-Particle Correlation Sensors** across the spacecraft in the formation provide coverage of multiple look-directions simultaneously (**Fig. 4**).

Option to examine magnetic correlation across phase-space by swapping **Electric Field Instruments** for **Search Coil Magnetometers** on some spacecraft.

4 CHARACTERISING ELECTRON-SCALE FLUCTUATIONS

Multipoint measurements of electromagnetic fields from two sensor types across the formation (**Fig. 4**) form **Distributed Wave Sensor**:

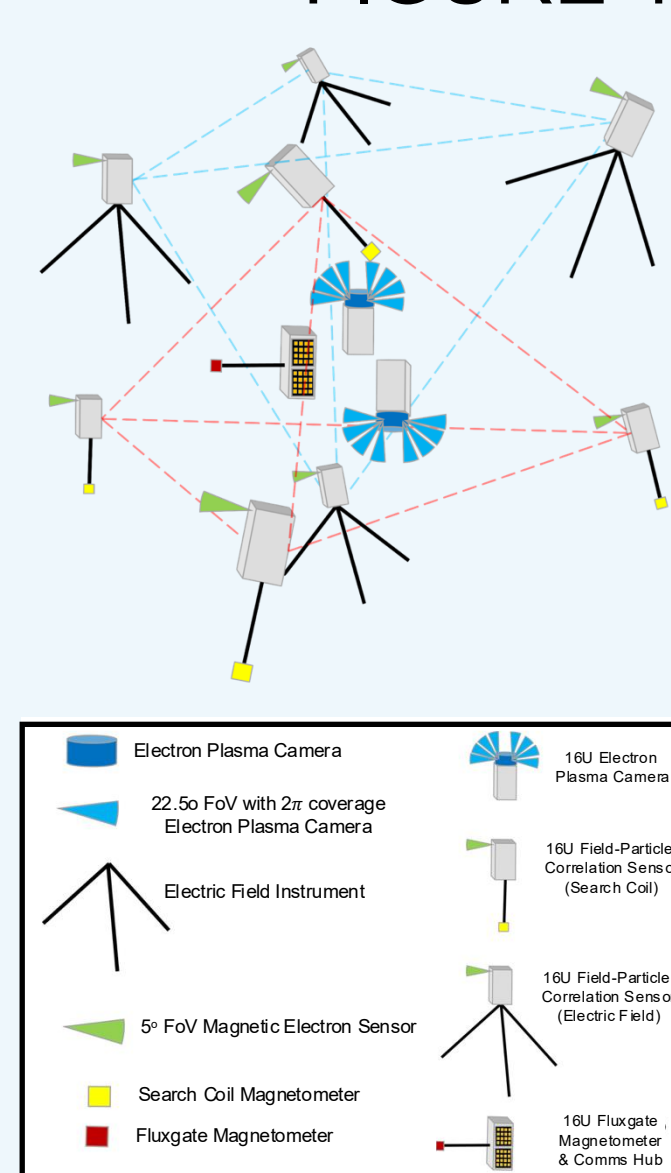
- Electric Filed Instrument** – 3D fluctuating electric field
- Search Coil Magnetometer** – 3D fluctuating magnetic field

Key Wave Properties: wavevector, phase speed, plasma-frame frequency

To target expected electron-scale fluctuations 3D spacecraft **separations from 0.5 – 10km** are required.

Could be achieved with large (11 s/c) swarm of CubeSats (**Fig. 4**) or small (4 s/c) formation of small satellites

FIGURE 4



Schematic of CubeSat swarm formulation of the mission

5 BACKGROUND PLASMA CONTEXT

Interpreting **Field-Particle Correlation Sensor** and **Distributed Wave Sensor** measurements requires contextual information about the plasma:

- Background magnetic field direction
- Characteristic electron length and time scales

Two sensor types provide necessary contextual measurements (**Fig. 4**):

- Fluxgate Magnetometer** – 3D DC magnetic field vector
- Electron Plasma Camera** – 3D electron distributions & moments [Morel+ (2017) *JGR* **122**]

Since these measurements are needed to characterise background plasma, they can be obtained at a single point in electron-scale formation